

Open book exam. Submit your answers to gradescope.

In your explanations and justifications, write complete and grammatically correct sentences.

Please do not ask questions during the exam.

6. Illustrate the application of divide and conquer to compute x^{256} for some number x .
Assuming p is a power of 2, how many multiplications are needed to compute x^p ?
Justify your answer, referring to the example for $p = 256$.

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Answer:

6. We apply repeated squaring.

(a) $z = x^2$, $z = z^2$ gives x^4 , $z = z^2$ gives x^8 , $z = z^2$ gives x^{16} , $z = z^2$ gives x^{32} , $z = z^2$ gives x^{64} , $z = z^2$ gives x^{128} , $z = z^2$ gives x^{256}

In the example we needed 8 multiplications, as $2^8 = 256$.

For any power p we need $\log_2(p)$ multiplications, following the recursion $T(p) = T(p/2) + 1$ similar to binary search, which resolves to $\log_2(p)$.